

Track 35: AutoCAD & Civil Design

June 2026

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Department: Civil Engineering | MR ROKESH TECHNOLOGY

Skill path: Drafting → Layers → Plans → Sections → Elevations → Site Layout → Quantity Estimation

Related: [Track 36 — BIM & Structural](#) | [Department Map](#)

Track Overview

Who this is for: Civil engineering students, draftsmen, site engineers, and architects entering construction design.

Prerequisites: Basic engineering drawing knowledge helpful; taught from fundamentals.

Tools: AutoCAD (latest LT/Full), Civil 3D (intro, 6-month+), DWG standards, plot/print setup.

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Drafting awareness	Floor plan + section drawing set
1 Month	80–100	Junior draftsman	Complete building drawing package
3 Months	250–300	Site engineer ready	Road layout + building project
6 Months	500–600	Professional draftsman	Portfolio of 5+ civil projects

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Interface, units, layers, line types, coordinates	Layer-standard template file
D3–4	2D drafting: walls, doors, windows, dimensions	Floor plan (residential)
D5–6	Sections, elevations, hatching, annotations	Section + elevation set
D7–8	Blocks, attributes, external references (XREF)	Reusable block library
D9–10	Plotting, scales, title blocks, capstone	Complete drawing set (PDF + DWG)

1-Month Foundation

Week	Topics	Project
W1	Precision drafting, polar tracking, constraints	Single-storey house plan
W2	Multi-storey plans, stair details, sections	G+1 building drawings
W3	Site plan, contour intro, road alignment basics	Site layout plan
W4	BOQ extraction intro, plotting standards	Building + site plan package

3-Month Job Ready

Month	Topics	Projects
M1	Advanced 2D, standards (CPWD/IS codes intro)	Residential building set
M2	Civil 3D intro: surfaces, alignments, profiles	Road design project
M3	Quantity estimation, capstone	Commercial building + road layout capstone

6-Month Professional

M1–M3 as above + M4: Civil 3D corridors + M5: Drainage/sewer layout + M6: Portfolio (5 projects) + estimation reports.

Assessment Rubric

Weekly Quiz 20% | Lab/drawing completion 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

Enforce layer naming standards from Week 1. All submissions in both DWG and PDF. Pair with [Track 36](#) for structural depth.